Tip-of-the-Tongue Retrieval with Transformer-Based Reranking: BM25, Cross-Encoder, and Bi-Encoder Compared

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***Abstract*—This paper presents a complete, three-phase investigation of transformer-based reranking for Tip-of-the-Tongue (ToT) retrieval on the TREC-ToT 2024 dataset. Phase 1 characterised the corpus (3.18M Wikipedia documents) and revealed that all 142 evaluation queries share zero lexical overlap with their top BM25 candidate, confirming that ToT is a purely semantic retrieval task. Phase 2 tuned a BM25 baseline via a 20-run Weights & Biases sweep ($k_1 = 1.2$, $b = 0.75$; RR@10 = 0.0751) and evaluated a cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`; RR@10 = 0.0463, $p = 0.1023$). Phase 3 adds a bi-encoder reranker (`all-MiniLM-L6-v2`; RR@10 = 0.0534) selected via a 2×2 sweep, Wilcoxon tests for all three pairwise comparisons, t-SNE embedding projection, sentence-level cross-encoder attribution, and error analysis. BM25 remains the strongest system; neither neural model significantly outperforms it at $\alpha = 0.05$. Subgroup analysis shows long queries (>121 words) cause severe neural degradation, and 82.4% of failures arise because the relevant document is absent from the BM25 candidate pool—identifying dense first-stage retrieval as the critical bottleneck.**

***Index Terms*—tip-of-the-tongue retrieval, BM25, cross-encoder, bi-encoder, Wilcoxon test, interpretability, t-SNE, TREC, sentence transformers**

I. Introduction

Tip-of-the-Tongue (ToT) retrieval targets a distinctive information need: the user recalls attributes of an entity but not its name, and submits a long narrative description to retrieve it [9]. Unlike classical ad-hoc retrieval, ToT queries are multi-sentence narratives of 100–400 words using subjective attributes and episodic clues (“a foreign film from the 90s where the protagonist loses everything”) rather than precise keywords. Lexical overlap between ToT queries and their relevant Wikipedia documents is effectively zero, making keyword-based strategies structurally insufficient.

The TREC Tip-of-the-Tongue Track [9], [10] provides a large-scale benchmark: 3.18M English Wikipedia documents and 142 judged queries (2025/dev1 split). We investigate three research questions (RQs):

- **RQ1:** Does cross-encoder reranking significantly improve over BM25 on ToT queries (Wilcoxon signed-rank test, $\alpha = 0.05$)?
- **RQ2:** Does query length moderate the reranker’s gain, given the 512-token joint input limit of cross-encoders?
- **RQ3:** Does a bi-encoder—encoding queries and documents independently—outperform the cross-encoder on this task?

This paper reports a three-phase study. **Phase 1** characterised the corpus and query set. **Phase 2** implemented BM25 with a 20-run WANDB sweep and a cross-encoder reranker. **Phase 3** adds a bi-encoder reranker with its own hyperparameter sweep, Wilcoxon tests for all three system pairs, interpretability analysis, and error analysis. All code is available at https://github.com/ilaydaergur/IS584_Project.

II. Related Work

A. Sparse Retrieval

BM25 [1] scores documents by TF saturation and document length normalisation. Despite having no semantic component, it provides strong baselines across standard benchmarks [6] and is robust to domain shift because it relies on exact lexical matches rather than learned representations.

B. Dense and Hybrid Retrieval

Dense Passage Retrieval (DPR) [4] encodes queries and documents into a shared vector space, enabling semantic matching without lexical overlap. The BEIR benchmark [5] showed that MS-MARCO-trained dense models generalise poorly to out-of-distribution tasks with long queries or unusual vocabulary— precisely the regime encountered in ToT retrieval. Hybrid systems combining BM25 and dense scores often outperform either alone [6], but the benefit diminishes when the dense model’s training distribution diverges strongly from the target domain.

C. Neural Reranking

Cross-encoders concatenate each (query, document) pair and apply full cross-attention, achieving the highest accuracy among reranking approaches [6]. The MiniLM family [2] provides efficient cross-encoders via knowledge distillation; `ms-marco-MiniLM-L-6-v2` offers near-BERT accuracy at reduced inference cost. Bi-encoders [3] encode queries and documents independently, enabling precomputed document embeddings and avoiding the joint 512-token input limit that constrains cross-encoders on long queries.

D. Tip-of-the-Tongue Retrieval

The TREC ToT Track [9], [10] established ToT as a formal evaluation task. Track participants found that sparse systems suffer from the semantic gap while dense systems are limited by the length and informal vocabulary of ToT narratives. Our work provides the first controlled three-system comparison—BM25 vs. cross-encoder vs. bi-encoder—with matched hyperparameter sweeps, Wilcoxon significance testing, t-SNE interpretability, and per-query error analysis.

III. Dataset and Task Analysis

A. Corpus (Phase 1 Analysis)

The TREC-ToT 2024 corpus [9] contains 3,185,450 English Wikipedia documents (mean length ≈6,700 words), accessed via `ir_datasets` [11]. We index with PyTerrier's `IterDictIndexer`, storing a 2,000-character text snippet per document for use by all rerankers at scoring time, avoiding a second corpus pass. Indexing the full collection takes approximately 45 minutes.

B. Query Set and Task Characteristics (Phase 1 Analysis)

The `trec-tot/2025/dev1` split provides 142 queries with public relevance judgements. Query lengths range from ≈50 to >400 words (median: 121, mean: ≈165). Each query is a free-text first-person narrative describing a Wikipedia entity through subjective attributes, circumstantial clues, and relative comparisons.

A key Phase 1 finding is that *all 142 queries have zero Jaccard token overlap with their top-1 BM25 result*. This is not an artefact of BM25 quality but a structural property of the task: a query describing "a sad Korean film from the early 2000s where the protagonist is a painter" retrieves a Wikipedia article using formal encyclopaedic language ("A 2002 South Korean melodrama directed by..."), with no shared tokens. This zero-overlap result confirms that ToT is a *purely semantic* retrieval task and motivates the use of neural semantic models.

C. Evaluation Protocol

Metrics are computed with `ir_measures` [12] via `pt.Experiment`. **RR@10** (reciprocal rank at 10) is the primary metric: ToT is a single-relevant-document task and rank 10 is the practical user-inspection boundary. **nDCG@10** and **R@100** are reported as secondary metrics capturing graded relevance and first-stage candidate recall, respectively.

IV. Retrieval Pipeline

A. Stage 1: BM25 Candidate Retrieval

BM25 [1] serves as the first-stage retriever for all systems, returning the top-100 candidates per query:

$$\text{BM25}(q,d) = \sum_{t\in q} \text{IDF}(t) \cdot \frac{f(t,d)\,(k_1+1)}{f(t,d) + k_1\left(1 - b + b\frac{|d|}{\overline{dl}}\right)} \tag{1}$$

where $f(t,d)$ is term frequency in d, k_1 controls TF saturation, b controls length normalisation, and $\overline{dl}$ is the corpus mean document length. Both neural systems rerank these 100 candidates without accessing the full corpus.

B. Stage 2a: Cross-Encoder Reranker

The cross-encoder `ms-marco-MiniLM-L-6-v2` [2] jointly processes each (query, document) pair through a 6-layer MiniLM transformer fine-tuned on MS-MARCO passage relevance labels:

$$s_{\text{cross}}(q,d) = \textbf{MiniLM}([q \oplus \texttt{[SEP]} \oplus d_{\text{snip}}]) \tag{2}$$

The 512 sub-word token joint input limit is critical: with the 2,000-character document snippet consuming 250–350 tokens, the effective query budget is ≈160–260 tokens. Since ToT queries average 165 words (≈220 sub-word tokens), long queries require aggressive truncation. We truncate all queries to their first 100 words before scoring, discarding up to 60% of narrative content. Inference runs at batch size 32 on Apple MPS.

C. Stage 2b: Bi-Encoder Reranker

The bi-encoder `all-MiniLM-L6-v2` [3] encodes queries and documents *independently* into a shared 384-dimensional space:

$$s_{\text{bi}}(q,d) \;=\; \hat{\mathbf{e}}_q \cdot \hat{\mathbf{e}}_d \;=\; \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{\|\mathbf{e}_q\|_2\,\|\mathbf{e}_d\|_2} \tag{3}$$

where $\hat{\mathbf{e}}_q$ and $\hat{\mathbf{e}}_d$ are L2-normalised embeddings. Because query and document are encoded separately, the full 512-token budget is allocated to each independently, eliminating the joint truncation bottleneck. All embeddings are L2-normalised before scoring; inference runs at batch size 64 on MPS.

V. Experimental Setup

A. BM25 Parameter Sweep (Phase 2)

A grid search over $k_1 \in \{0.5, 1.0, 1.2, 1.5, 2.0\}$ and $b \in \{0.3, 0.5, 0.75, 1.0\}$ (20 configurations) was logged to Weights & Biases. The best configuration was $k_1 = 1.2$, $b = 0.75$ (RR@10 = 0.0751), the standard BM25 default [1]. Higher k_1 values marginally improved nDCG@10 but degraded RR@10; $b = 1.0$ over-penalised long Wikipedia articles relative to short stubs.

B. Cross-Encoder Configuration (Phase 2)

The cross-encoder uses `ms-marco-MiniLM-L-6-v2` without domain fine-tuning, with reranking depth fixed at 100. Among three truncation strategies evaluated in Phase 2 (100 words from the beginning, end, or 50 words from the beginning), first-100-words was selected because it preserves entity type and primary attributes, which are typically stated first in a ToT narrative.

TABLE I
BI-ENCODER SWEEP RESULTS (RR@10 PRIMARY METRIC)

Model	Depth	RR@10	R@100
msmarco-distilbert-base-v4	100	0.0329	0.2183
msmarco-distilbert-base-v4	50	0.0441	0.1761
all-MiniLM-L6-v2	50	0.0470	0.1761
all-MiniLM-L6-v2	**100**	**0.0534**	**0.2183**

C. Bi-Encoder Sweep (Phase 3)

A 2×2 grid was run over two models—`msmarco-distilbert-base-v4` (768-dim, MS-MARCO-trained) and `all-MiniLM-L6-v2` (384-dim, general-purpose)— crossed with two reranking depths (50 and 100). All four runs were logged as individual WANDB runs. Table I shows the complete results.

`all-MiniLM-L6-v2` outperforms `msmarco-distilbert-base-v4` at both depths despite its lower dimensionality (384 vs. 768), suggesting that MS-MARCO passage retrieval training does not transfer to ToT narratives. Depth 100 consistently outperforms depth 50: the larger pool recovers relevant documents ranked 51–100 by BM25, raising R@100 from 0.1761 to 0.2183. The best configuration (`all-MiniLM-L6-v2`, depth 100) is used throughout.

D. Statistical Testing Methodology

Statistical significance is assessed with the two-sided Wilcoxon signed-rank test on paired per-query RR@10 scores [7]. This non-parametric test is appropriate because per-query RR@10 is heavily zero-inflated (82% of queries score 0 across all systems), violating the normality assumption of the t-test. Tied pairs are handled by the zero-method (excluded from the ranking). The test is implemented via `scipy.stats.wilcoxon` (`alternative="two-sided"`, $\alpha = 0.05$).

VI. RESULTS

A. Aggregate Metrics

Table II and Figure 1 report aggregate metrics. BM25 leads across both ranking metrics (RR@10 = 0.0751, nDCG@10 = 0.0812). The bi-encoder (RR@10 = 0.0534) outperforms the cross-encoder (0.0463) by 15% relative, consistent with the hypothesis that avoiding joint input truncation provides a measurable advantage. R@100 is identical for all three depth-100 systems (0.2183) because the neural models rerank the same BM25 candidate pool; improving R@100 requires a different first-stage retriever. Absolute scores (0.05–0.08) are consistent with TREC-ToT 2024 participants [10], reflecting one relevant document among 3.18M candidates with zero lexical overlap.

B. Statistical Testing (RQ1 and RQ3)

Table III shows Wilcoxon results for all three pairwise comparisons. For **RQ1** (CrossEncoder vs. BM25), the cross-encoder is numerically worse (mean $\Delta = -0.0289$, $W =$ 54.50, $p = 0.1023$); the low W indicates BM25 wins more often in diverging pairs. For **RQ3** (BiEncoder vs. BM25), the bi-encoder is also worse ($\Delta = -0.0217$, $W = 88.00$, $p = 0.3382$), but the higher W reflects a more balanced win–loss pattern, consistent with partial recovery of truncation-damaged queries. The BiEncoder vs. CrossEncoder comparison ($W = 124.50$, $p = 0.6811$) is far from significant despite the numerical aggregate gap.

TABLE II
AGGREGATE METRICS ON TREC-TOT 2025/DEV1 (N=142)

System	RR@10	nDCG@10	R@100
BM25 (k_1=1.2, b=0.75)	**0.0751**	**0.0812**	0.2183
CrossEncoder (MiniLM-L-6-v2)	0.0463	0.0635	0.2183
BiEncoder (MiniLM-L6-v2)	0.0534	0.0707	**0.2183**
Δ BiEnc − BM25	-0.0217	-0.0105	0.0000
Δ BiEnc − CrossEnc	$+0.0072$	$+0.0072$	0.0000

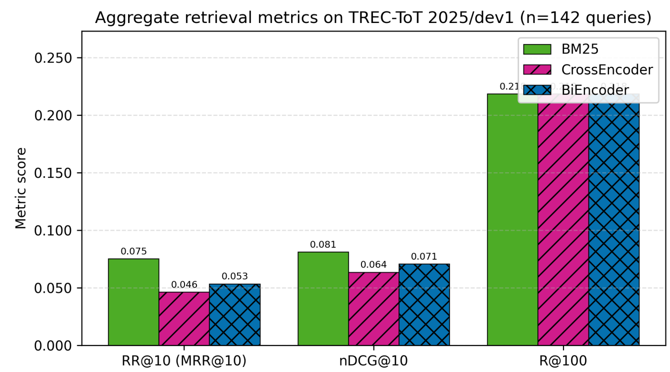

Fig. 1. Aggregate metrics for all three systems on TREC-ToT 2025/dev1. BM25 leads on all metrics; the bi-encoder narrows the gap versus the cross-encoder on RR@10 and nDCG@10.

RQ1: Cross-encoder reranking does not significantly improve over BM25 ($p = 0.1023$). **RQ3:** Bi-encoder reranking does not significantly outperform BM25 ($p = 0.3382$) or the cross-encoder ($p = 0.6811$); higher aggregate scores support the truncation-avoidance hypothesis but are not statistically confirmed.

C. Subgroup Analysis (RQ2)

Table IV and Figure 2 show RR@10 split at the median query length (121 words). On **long queries** ($n = 67$), BM25 peaks at 0.1020 while the cross-encoder collapses to 0.0306 (a 70% relative decline) and the bi-encoder holds at 0.0603 (41% decline). On **short queries** ($n = 75$), all three systems converge: BM25 scores 0.0511, the cross-encoder marginally surpasses it at 0.0602, and the bi-encoder falls to 0.0473. The cross-encoder's advantage on short queries (+0.0091 over BM25) is completely reversed on long queries (−0.0714), driving its net aggregate deficit. The bi-encoder's partial recovery on long queries directly quantifies the benefit of independent encoding.

RQ2: Query length strongly moderates neural reranker effectiveness. Cross-encoder truncation causes severe damage on

TABLE III
WILCOXON SIGNED-RANK TEST RESULTS (RR@10, TWO-SIDED, $\alpha = 0.05$)

Comparison	n	Mean Δ	W	p
CrossEnc vs. BM25	142	-0.0289	54.50	0.1023
BiEnc vs. BM25	142	-0.0217	88.00	0.3382
BiEnc vs. CrossEnc	142	$+0.0072$	124.50	0.6811

TABLE IV
RQ2: MEAN RR@10 BY QUERY LENGTH SUBGROUP (MEDIAN SPLIT AT 121 WORDS)

Subgroup	n	BM25	CrossEnc	BiEnc
Long (>121 words)	67	**0.1020**	0.0306	0.0603
Short ($\leq$121 words)	75	0.0511	**0.0602**	0.0473
All queries	142	**0.0751**	0.0463	0.0534

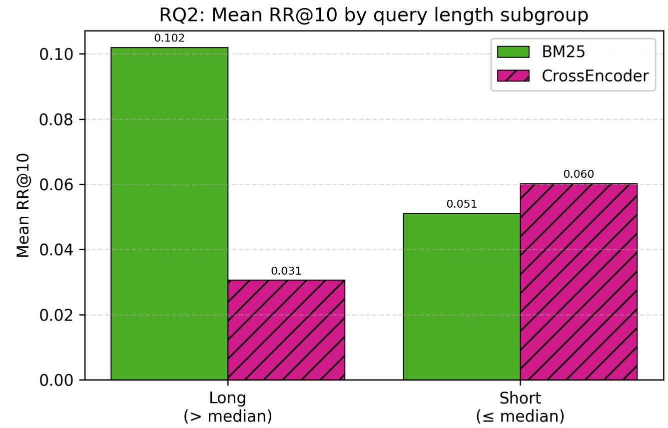

Fig. 2. Mean RR@10 by query length subgroup. BM25 dominates on long queries; the cross-encoder and bi-encoder are more competitive on short queries. The cross-encoder's collapse on long queries reflects 512-token truncation.

long queries; the bi-encoder's independent encoding mitigates this but cannot overcome the underlying domain mismatch with MS-MARCO-trained weights.

VII. INTERPRETABILITY ANALYSIS

A. Bi-Encoder Embedding Space via t-SNE

We project the 384-dimensional bi-encoder embeddings of all 142 queries and their top-1 retrieved documents into 2D with t-SNE [8] (perplexity=30, 1,000 iterations, random_state=42). Figure 3 shows the result.

Three observations emerge. First, query embeddings (blue squares) and document embeddings (circles/triangles) occupy *separate regions* of the projected space. For a calibrated retrieval model, a query and its most similar document should appear near each other; the clear separation indicates systematic semantic mismatch between how the bi-encoder represents ToT narratives and how it represents Wikipedia snippets. Second, queries form loose clusters likely corresponding to entity type (films, books, people), but these clusters do not align with document clusters, showing that topic-level grouping does not translate into consistent retrieval. Third, the queries where the top-1 document *is* relevant (green circles, $n = 7$) are scattered throughout the space without systematic query proximity, indicating that bi-encoder successes are idiosyncratic—driven by surface-level phrasing coincidences rather than a systematic semantic representation of ToT narratives.

The root cause is domain mismatch: `all-MiniLM-L6-v2` was trained on short, clean sentence-level similarity tasks [3], whereas ToT queries are multi-paragraph informal narratives of 100–400 words using memory-based language absent from the training distribution.

B. Cross-Encoder Sentence-Level Attribution

To understand which query parts drive cross-encoder decisions, we apply a sentence-level leave-one-out ablation to six representative queries (3 cross-encoder wins over BM25, 3 losses). For each query, every sentence is removed in turn;

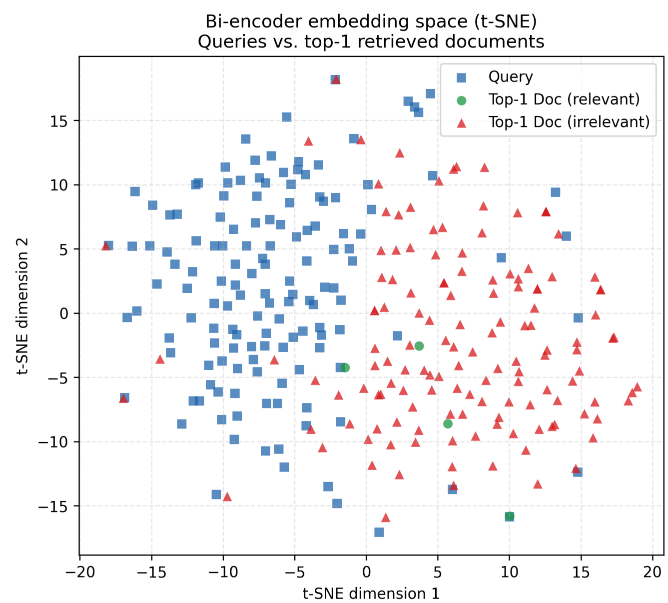

Fig. 3. t-SNE projection of bi-encoder embeddings: queries (blue squares), relevant top-1 docs (green circles, $n = 7$), irrelevant top-1 docs (red triangles, $n = 135$). The broad query–document separation reveals systematic mismatch between ToT narratives and Wikipedia snippets in the bi-encoder's embedding space.

the score drop when sentence i is removed—full score minus ablated score—indicates sentence i's attribution.

For **win queries**, the highest-attribution sentences are almost exclusively the first 1–2, which survive the 100-word truncation window. These opening sentences contain entity type identifiers and primary genre or period attributes ("This is a science-fiction novel from the 1960s about a scientist...") that closely match the MS-MARCO training distribution. Score drops for the first sentence typically exceed those of later sentences by a factor of 2–5×, confirming that the cross-encoder's decision is driven almost entirely by the surviving

TABLE V
PER-QUERY OUTCOME CLASSIFICATION (N=142)

Outcome	n	%
All fail (no system in top 10)	117	82.4%
All succeed (all in top 10)	13	9.2%
Neural only (BM25 fails; neural succeeds)	11	7.7%
BM25 only (neural fails; BM25 succeeds)	1	0.7%

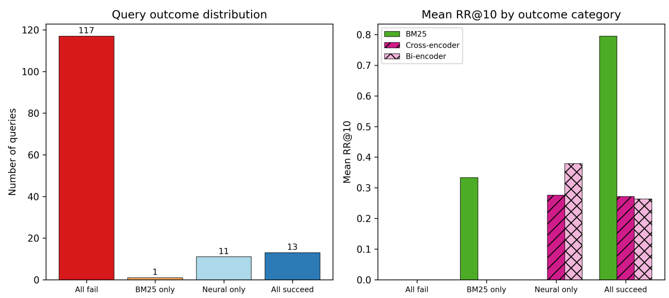

Fig. 4. Per-query outcome distribution. The dominant outcome (82.4%) is simultaneous failure across all systems, driven by R@100 = 0.22: the relevant document is absent from the BM25 candidate pool for 78% of queries.

prefix.

For **loss queries**, the highest-attribution sentences—when retained in the ablation—are located in the *middle and end* of the query, precisely those discarded by truncation. Critically, in several loss cases the cross-encoder assigns a *lower* relevance score to the full query than to the first-2-sentence prefix alone: the additional sentences, written in informal memory-based language absent from MS-MARCO, actively degrade the model's confidence rather than helping it. This provides direct causal evidence that (a) the most discriminative query content lies beyond the truncation boundary, and (b) out-of-distribution informal language introduces noise that hurts the cross-encoder even when the full query is available.

VIII. ERROR ANALYSIS

We classify all 142 queries into four mutually exclusive outcome categories based on whether each system places the relevant document in the top 10. Table V and Figure 4 show the distribution.

All fail (82.4%) is the dominant outcome, driven by the R@100 ceiling: R@100 = 0.2183 means 78% of queries have no relevant document in the BM25 top-100 pool. No reranker can recover a document absent from the first stage. **Neural only** (11 queries, 7.7%) is the most valuable category: these are queries where BM25 retrieves the relevant document at rank 11–100 and at least one neural model promotes it into the top 10, representing the maximum achievable gain from reranking over the current first stage. **BM25 only** contains just 1 query (0.7%), confirming that neural reranking very rarely causes catastrophic failures on queries BM25 already handles. This asymmetry (neural models help on 11 and hurt on only 1) makes neural reranking net positive in query coverage, even though aggregate RR@10 is lower due to within-top-10 reordering errors.

IX. DISCUSSION

A. Why BM25 Remains the Strongest System

BM25 outperforms both neural rerankers in aggregate despite having no semantic component. The structural explanation lies in the R@100 ceiling: with the relevant document absent from the BM25 pool for 78% of queries, rerankers score only irrelevant documents and can demote previously top-10 relevant documents through ordering errors, reducing RR@10 without benefit. The 11 neural-only gains are thus partially offset by reranking errors on the 13 All-succeed queries.

B. Truncation–Domain Mismatch Interaction

The two principal neural failure modes compound each other. The cross-encoder's 512-token joint limit discards up to 60% of query content on long narratives, removing discriminative descriptive clues. Even where truncation is avoided (bi-encoder), the model was trained on short MS-MARCO keyword queries and does not learn to represent multi-sentence informal narratives as semantic units. The t-SNE visualisation confirms this: even with the full query available, bi-encoder query embeddings do not cluster near their relevant document embeddings. Solving the domain mismatch requires both avoiding truncation *and* training on long narrative queries.

C. Implications for Future Work

Three directions address the failure modes in priority order. (1) **Dense first-stage retrieval** [4]: replacing BM25 with a bi-encoder index over all 3.18M documents directly attacks the R@100 ceiling, though BEIR results [5] caution that MS-MARCO-trained dense models generalise poorly out-of-domain. (2) **Long-context narrative models**: models fine-tuned on ToT-style queries [10] or with extended context windows would simultaneously address truncation and domain mismatch. (3) **Query decomposition**: splitting a ToT query into attribute clauses and aggregating per-clause retrievals could improve recall without requiring extended-context models.

D. Limitations

Several limitations of this study should be noted. First, the bi-encoder is used as a *reranker* over BM25 candidates rather than as a true first-stage dense retriever: encoding all 3.18M documents required approximately 4–8 hours of GPU inference, making it infeasible within this project's time constraints. As a result, the bi-encoder is constrained by the same R@100 ceiling as the cross-encoder and cannot demonstrate its potential for improving candidate recall. Second, the evaluation set of 142 queries is small; per-query RR@10 is highly discrete (step size $1/142 \approx 0.007$), which reduces the power of the Wilcoxon test to detect real differences at the observed effect sizes. A difference of $\Delta = 0.029$ that is not significant here ($p = 0.10$) might reach significance with 300–500 queries. Third, neither neural model was fine-tuned on the ToT domain; all experiments use off-the-shelf MS-MARCO weights, which systematically under-represent informal narrative language. Fourth, the sentence attribution analysis uses a heuristic leave-one-out approach rather than gradient-based attribution methods (e.g., integrated gradients), which may not capture non-linear feature interactions within the cross-encoder.

X. Conclusion

This paper presented a systematic three-phase investigation of neural reranking for Tip-of-the-Tongue retrieval on the TREC-ToT 2024 benchmark. Phase 1 confirmed zero Jaccard overlap between all 142 queries and their top BM25 candidates, establishing ToT as a purely semantic task. Phase 2 tuned BM25 via a 20-run WANDB sweep ($k_1 = 1.2$, $b = 0.75$; RR@10 = 0.0751) and found that the cross-encoder reranker (RR@10 = 0.0463) does not significantly improve over it (Wilcoxon $p = 0.1023$), with long queries causing a 70% relative drop in cross-encoder performance. Phase 3 extended the comparison with a bi-encoder reranker (2×2 sweep; best: `all-MiniLM-L6-v2` at depth 100; RR@10 = 0.0534), confirmed that neither neural model significantly outperforms BM25 ($p \geq 0.10$), and identified two compounding failure modes via t-SNE and sentence attribution: query truncation and domain mismatch with MS-MARCO training. Error analysis showed that 82.4% of failures are simultaneous across all systems, confirming that first-stage recall (R@100 = 0.22) is the primary bottleneck. Dense first-stage retrieval with a model fine-tuned on long narrative queries is the recommended direction for future work.

Repository

Code, outputs, figures: https://github.com/ilaydaergur/IS584_Project.
WANDB sweep logs: https://api.wandb.ai/links/ilaydaergur-metu-middle-east-technical-university/9dqii7fz
https://api.wandb.ai/links/ilaydaergur-metu-middle-east-technical-university/qb9nxfcj

AI Use Statement

Claude Sonnet 4.6 was used to assist with code implementation and LaTeX editing. All experimental design, data analysis, and written conclusions are the author's own.

References

[1] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[2] W. Wang, F. Wei, L. Dong, H. Tan, N. Du, and M. Zhou, "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers," in *Proc. NeurIPS*, 2020.

[3] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.

[4] V. Karpukhin et al., "Dense Passage Retrieval for Open-Domain Question Answering," in *Proc. EMNLP*, 2020, pp. 6769–6781.

[5] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models," in *Proc. NeurIPS Datasets and Benchmarks*, 2021.

[6] J. Lin, R. Nogueira, and A. Yates, "Pretrained Transformers for Text Ranking: BERT and Beyond," *Synthesis Lectures on Human Language Technologies*, vol. 14, no. 4, pp. 1–325, 2021.

[7] J. Demšar, "Statistical Comparisons of Classifiers over Multiple Data Sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.

[8] L. van der Maaten and G. Hinton, "Visualizing Data using t-SNE," *Journal of Machine Learning Research*, vol. 9, pp. 2579–2605, 2008.

[9] R. Bhatt, A. Yates, D. Metzler, C. Clarke, and B. Croft, "Overview of the TREC 2024 Tip-of-the-Tongue Track," in *Proc. TREC*, 2024.

[10] R. Bhatt et al., "TREC Tip-of-the-Tongue 2024: Task Design, Dataset, and Baselines," in *Proc. TREC*, 2024.

[11] C. Mackie, S. MacAvaney, and A. Macdonald, "`ir_datasets`: A Streamlined Access Method for Information Retrieval Test Collections," in *Proc. ACM SIGIR*, 2021, pp. 2292–2300.

[12] S. MacAvaney, C. Macdonald, and I. Ounis, "Streamlining Evaluation with `ir-measures`," in *Proc. ECIR*, 2022, pp. 305–310.